UNIT – III

1. Define an autoencoder.
2. List the main components of an autoencoder architecture.
3. What is the role of the encoder and decoder in an autoencoder?
4. Define sparse coding.
5. What is a denoising autoencoder?
6. What is a Generative Adversarial Network (GAN)?
7. Name the two networks involved in GAN architecture.
8. List any four applications of autoencoders.
9. Explain the architecture of an autoencoder with a neat diagram.
10. Describe the training process of an autoencoder.
11. Explain undercomplete autoencoders and their advantages.
12. Discuss regularized autoencoders with examples.
13. Explain stochastic encoders and decoders.
14. Describe the working principle of a denoising autoencoder.
15. Explain the basic GAN framework and its objective function.
16. Discuss the difference between generative and discriminative models.
17. Explain contractive autoencoders and how they improve robustness.
18. Discuss the challenges in training GANs such as mode collapse and oscillation loss.
19. Explain how Wasserstein GAN improves GAN training stability.
20. Describe CycleGAN architecture and its applications.
21. Explain neural style transfer using deep generative models.

22. An autoencoder compresses a 100-dimensional input into a 20-dimensional latent space.

- Identify whether it is overcomplete or undercomplete.
- Justify your answer.

23. Given a reconstruction loss function

$$L = \|X - \hat{X}\|^2$$

If $X = [1, 2, 3]$ and $\hat{X} = [0.9, 2.1, 2.8]$, compute the reconstruction loss.

24. A GAN discriminator outputs a probability of 0.85 for real data and 0.30 for generated data.

- Interpret these results in terms of GAN training.

25. If the learning rate of a GAN generator is increased excessively, explain numerically and conceptually how it may affect convergence.

Answer key - numerical

Q22. Autoencoder dimensionality

Given:

Input dimension = 100

Latent space dimension = 20

Solution:

Since the latent space dimension is **less than** the input dimension:

$$20 < 100$$

This autoencoder is an **Undercomplete Autoencoder**.

Justification:

Undercomplete autoencoders enforce compression, forcing the model to learn the most important features and avoid trivial identity mapping.

Answer: Undercomplete Autoencoder.

Q23. Reconstruction Loss Calculation

Given:

$$X = [1, 2, 3]$$

$$\hat{X} = [0.9, 2.1, 2.8]$$

Reconstruction loss:

$$L = \|X - \hat{X}\|^2$$

Step 1: Compute difference

$$X - \hat{X} = [1 - 0.9,\ 2 - 2.1,\ 3 - 2.8] = [0.1,\ -0.1,\ 0.2]$$

Step 2: Square each term

$$[0.01,\ 0.01,\ 0.04]$$

Step 3: Sum

$$L = 0.01 + 0.01 + 0.04 = 0.06$$

Answer:

Q24. GAN Discriminator Output Interpretation

Given:

- Discriminator output for real data = **0.85**
- Discriminator output for generated data = **0.30**

Interpretation:

- The discriminator correctly classifies real data with high confidence.
- Generated data is poorly fooling the discriminator.
- Generator performance is weak and needs improvement.

Conclusion:

The GAN is **not yet well trained**; the generator must improve to produce more realistic samples.

Q25. Effect of Excessive Learning Rate in GAN

Conceptual-Numerical Explanation:

If learning rate α is very large:

- Weight updates become large:

$$W_{new} = W_{old} - \alpha \nabla L$$

- This can cause:
  - Overshooting the minimum
  - Oscillatory behavior
  - Training instability
  - Non-convergence

Conclusion:

Excessively high learning rate leads to **divergence and unstable GAN training**.

UNIT – IV

1. Define novelty detection.
2. What is outlier detection?

3. Differentiate between novelty detection and outlier detection.
4. List any four applications of outlier detection.
5. What is density estimation?
6. Define histogram-based density estimation.
7. What is a Kohonen Self-Organizing Map (SOM)?
8. What is a Restricted Boltzmann Machine (RBM)?
9. Explain different methods of outlier detection.
10. Describe kernel density estimation (KDE).
11. Compare histogram density estimation and KDE.
12. Explain the architecture of Kohonen Self-Organizing Feature Maps.
13. Describe the training algorithm of SOM.
14. Explain Kohonen Self-Organizing Motor Map.
15. Explain the structure of a Restricted Boltzmann Machine.
16. Discuss applications of RBMs in unsupervised learning.
17. Explain how novelty detection is applied in anomaly detection systems.
18. Analyze the role of neighborhood functions in SOM training.
19. Explain contrastive divergence learning in RBMs.
20. Discuss limitations of SOM and RBMs.
21. Compare SOM and RBM in terms of learning and representation.
22. Given data points: {2, 4, 4, 4, 6, 8}, construct a histogram with bin width = 2.
23. Using kernel density estimation, explain how bandwidth selection affects density smoothness.
24. In a SOM with learning rate 0.5 and weight vector

$$W = [0.2, 0.6]$$

Input vector

$$X = [0.6, 0.8]$$

Compute the updated weight vector after one iteration.

25. An RBM has 6 visible units and 3 hidden units.
    - Calculate the total number of weights (excluding biases).

Q22. Histogram Construction

Given Data:

$$\{2, 4, 4, 4, 6, 8\}$$

Bin width = 2

Bins:

- 2–4
- 4–6
- 6–8
- 8–10

Frequency Count:

Bin Range	Frequency
2–4	1
4–6	3
6–8	1
8–10	1

Answer: Histogram constructed with the above frequencies.

Q23. Bandwidth Effect in Kernel Density Estimation

Explanation:

- **Small bandwidth:**
  - Sharp peaks
  - Overfitting
- **Large bandwidth:**
  - Smooth curve
  - Underfitting

Conclusion:

Bandwidth controls the **bias–variance tradeoff** in KDE.

Q24. SOM Weight Update

Given:

Learning rate $\eta = 0.5$

Initial weight vector:

$$W = [0.2, 0.6]$$

Input vector:

$$X = [0.6, 0.8]$$

Update formula:

$$W_{new} = W + \eta(X - W)$$

Step 1: Compute $X - W$

$$[0.6 - 0.2,\ 0.8 - 0.6] = [0.4,\ 0.2]$$

Step 2: Multiply by learning rate

$$0.5 \times [0.4,\ 0.2] = [0.2,\ 0.1]$$

Step 3: Update weights

$$W_{new} = [0.2,\ 0.6] + [0.2,\ 0.1] = [0.4,\ 0.7]$$

Answer:

$$\boxed{W_{new} = [0.4,\ 0.7]}$$

Q25. RBM Weight Calculation

Given:

Visible units = 6

Hidden units = 3

Formula:

$$\text{Total weights} = \text{Visible} \times \text{Hidden}$$

Calculation:

$$6 \times 3 = 18$$

Answer:

$$\boxed{18 \text{ weights}}$$